

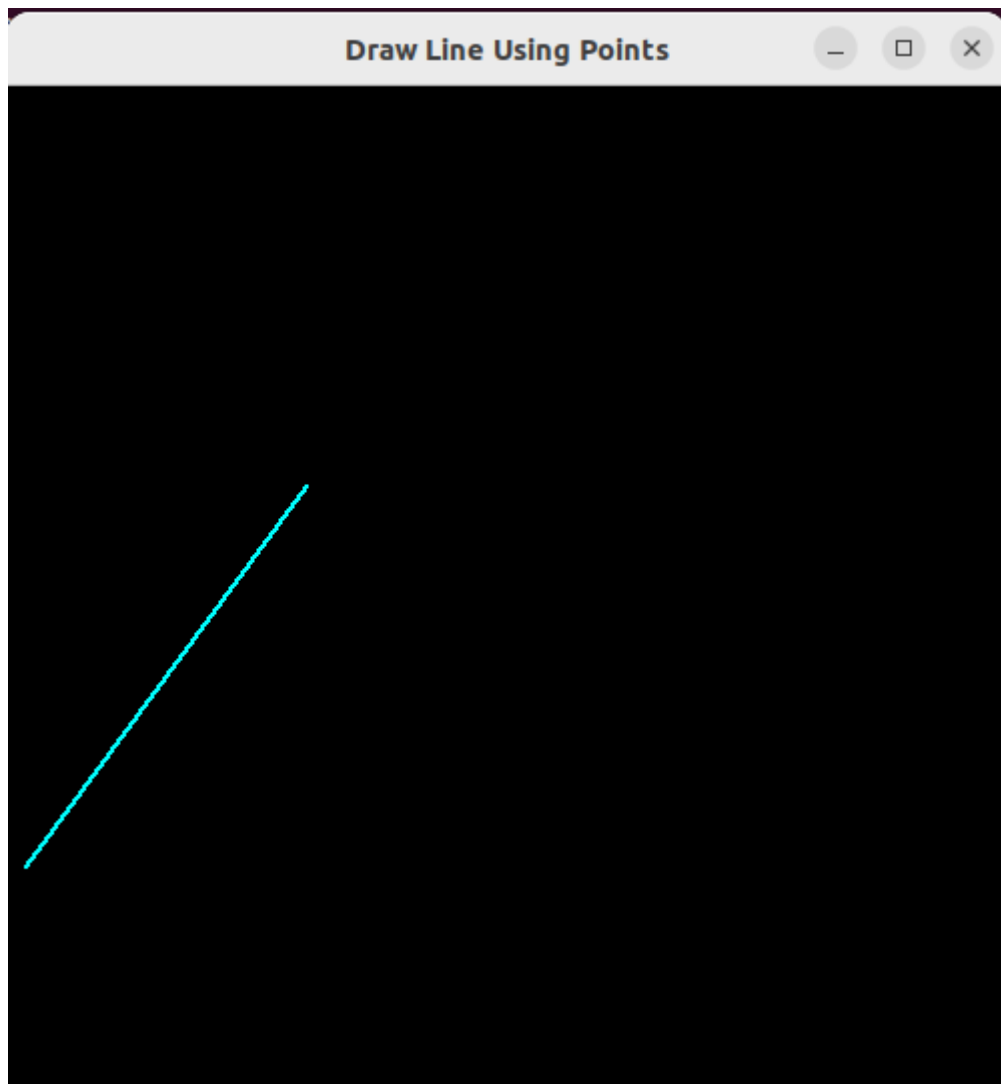
1. DDA Algorithm

```
1 #include <GL/glut.h>
2 #include <iostream>
3 #include <cmath>
4 using namespace std;
5
6 double x1 = 0.0, Y1 = 0.0;
7 double x2 = 0.0, Y2 = 0.0;
8
9 void init()
10 {
11     glClearColor(0.0, 0.0, 0.0, 0.0);
12     glMatrixMode(GL_PROJECTION);
13     gluOrtho2D(0.0, 500.0, 0.0, 500.0);
14 }
15
16 void input()
17 {
18     cout << "Enter the coordinates of the first point: ";
19     cin >> x1 >> Y1;
20     cout << "Enter the coordinates of the second point: ";
21     cin >> x2 >> Y2;
22 }
23
24 void display()
25 {
26     glClear(GL_COLOR_BUFFER_BIT);
27     glColor3f(0.0, 1.0, 1.0);
28     glPointSize(2.0);
29
30     double x = x1;
31     double y = Y1;
32
33     double dx = x2 - x1;
34     double dy = Y2 - Y1;
35
36     double slope = fabs(dy / dx);
37
38     int xStep = (dx >= 0) ? 1 : -1;
39     int yStep = (dy >= 0) ? 1 : -1;
40
41     glBegin(GL_POINTS);
42 }
```

```

43     if (slope <= 1)
44     {
45         while ((xStep > 0 && x <= x2) || (xStep < 0 && x >= x2))
46         {
47             glVertex2i((int)round(x), (int)round(y));
48             x += xStep;
49             y += slope * yStep;
50         }
51     }
52     else
53     {
54         while ((yStep > 0 && y <= Y2) || (yStep < 0 && y >= Y2))
55         {
56             glVertex2i((int)round(x), (int)round(y));
57             y += yStep;
58             x += (1.0 / slope) * xStep;
59         }
60     }
61
62     glEnd();
63     glFlush();
64 }
65
66
67 int main(int argc, char **argv)
68 {
69     input();
70     glutInit(&argc, argv);
71     glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
72     glutInitWindowSize(500, 500);
73     glutCreateWindow("Draw Line Using Points");
74     init();
75     glutDisplayFunc(display);
76     glutMainLoop();
77     return 0;
78 }
79

```



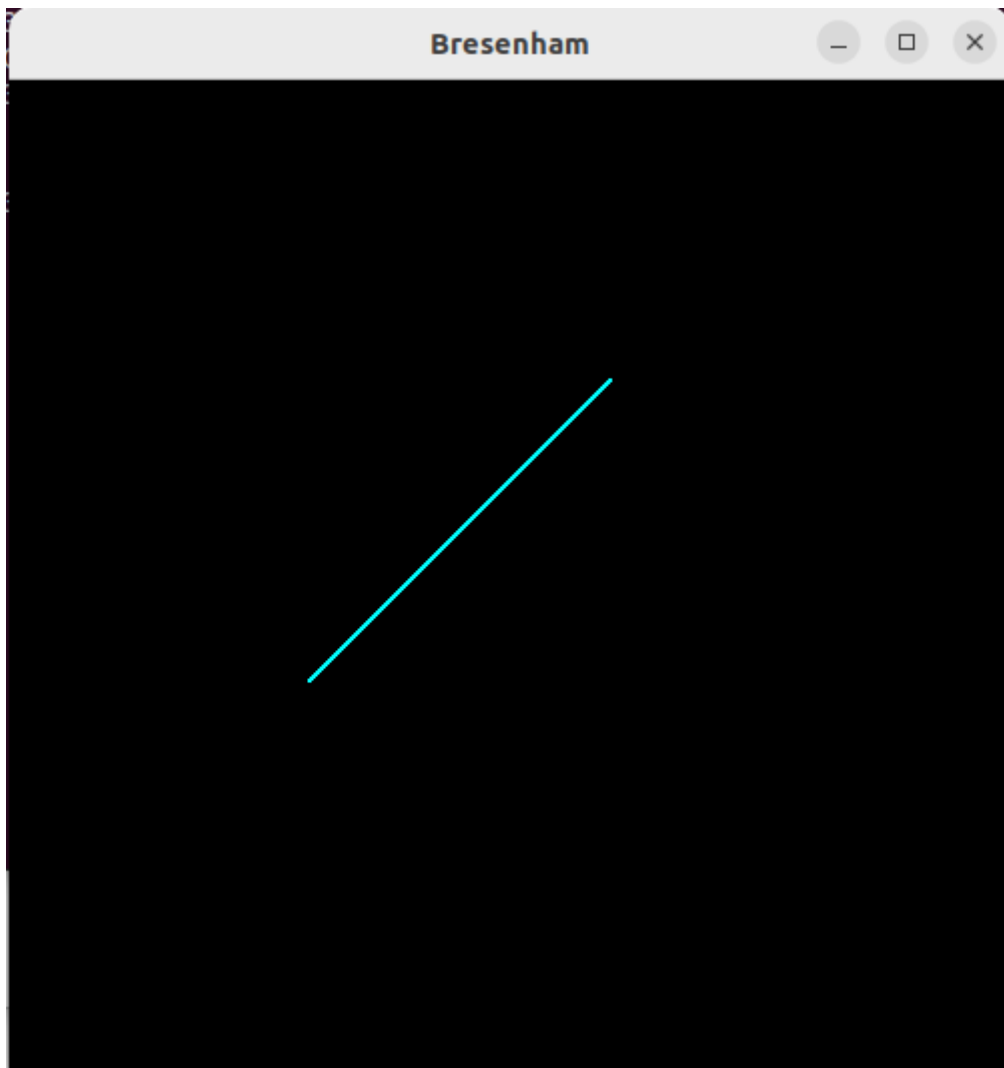
2. Bresenham's Line Drawing Algorithm

```
1 #include <GL/glut.h>
2 #include <iostream>
3 #include <cmath>
4 using namespace std;
5
6 double x1 = 0.0, Y1 = 0.0;
7 double x2 = 0.0, Y2 = 0.0;
8
9 void init()
10 {
11     glClearColor(0.0, 0.0, 0.0, 0.0);
12     glMatrixMode(GL_PROJECTION);
13     gluOrtho2D(0.0, 500.0, 0.0, 500.0);
14 }
15
16 void input()
17 {
18     cout << "Enter the coordinates of the first point: ";
19     cin >> x1 >> Y1;
20     cout << "Enter the coordinates of the second point: ";
21     cin >> x2 >> Y2;
22 }
23
24 void display()
25 {
26     glClear(GL_COLOR_BUFFER_BIT);
27     glColor3f(0.0, 1.0, 1.0);
28     glPointSize(2.0);
29
30     double dy = Y2 - Y1;
31     double dx = x2 - x1;
32     double slope = (dy * 1.0 / dx);
33
34     double p = (2 * dy) - dx;
35     double x = x1;
36     double y = Y1;
37     glBegin(GL_POINTS);
```

```

38     while(x <= x2)
39     {
40         glVertex2i(x, y);
41         if(p < 0)
42         {
43             p += 2 * dy;
44         }
45         else
46         {
47             y += 1;
48             p += 2 * dy - 2 * dx;
49         }
50         x += 1;
51     }
52
53     glEnd();
54     glFlush();
55 }
56
57
58 int main(int argc, char **argv)
59 {
60     input();
61     glutInit(&argc, argv);
62     glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
63     glutInitWindowSize(500, 500);
64     glutCreateWindow("Draw Line Using Points");
65     init();
66     glutDisplayFunc(display);
67     glutMainLoop();
68     return 0;
69 }
70

```



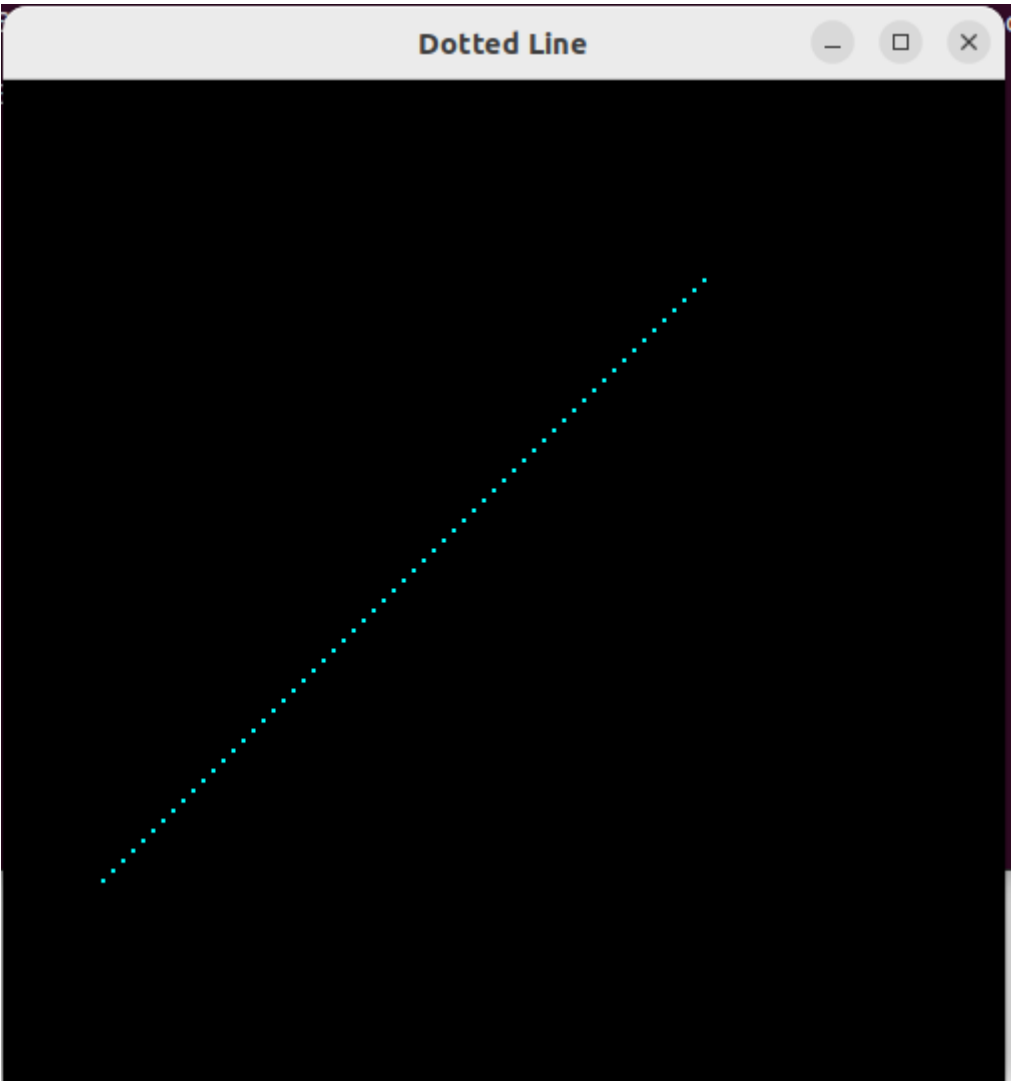
3. Dotted Line

```
1 #include <GL/glut.h>
2 #include <iostream>
3 #include <cmath>
4 using namespace std;
5
6 double x1 = 0.0, Y1 = 0.0;
7 double x2 = 0.0, Y2 = 0.0;
8
9 void init()
10 {
11     glClearColor(0.0, 0.0, 0.0, 0.0);
12     glMatrixMode(GL_PROJECTION);
13     gluOrtho2D(0.0, 500.0, 0.0, 500.0);
14 }
15
16 void input()
17 {
18     cout << "Enter the coordinates of the first point: ";
19     cin >> x1 >> Y1;
20     cout << "Enter the coordinates of the second point: ";
21     cin >> x2 >> Y2;
22 }
23
24 void display()
25 {
26     glClear(GL_COLOR_BUFFER_BIT);
27     glColor3f(0.0, 1.0, 1.0);
28     glPointSize(2.0);
29
30     double dy = Y2 - Y1;
31     double dx = x2 - x1;
32     double slope = (dy * 1.0 / dx);
33
34     double p = (2 * dy) - dx;
35     double x = x1;
36     double y = Y1;
37     int count = 0;
```

```

38     glBegin(GL_POINTS);
39     while(x <= x2)
40     {
41         if(count == 0)
42         {
43             glVertex2i(x, y);
44             count = 5;
45         }
46         count--;
47         if(p < 0)
48         {
49             p += 2 * dy;
50         }
51         else
52         {
53             y += 1;
54             p += 2 * dy - 2 * dx;
55         }
56         x += 1;
57     }
58
59     glEnd();
60     glFlush();
61 }
62
63
64 int main(int argc, char **argv)
65 {
66     input();
67     glutInit(&argc, argv);
68     glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
69     glutInitWindowSize(500, 500);
70     glutCreateWindow("Draw Line Using Points");
71     init();
72     glutDisplayFunc(display);
73     glutMainLoop();
74     return 0;
75 }
76

```



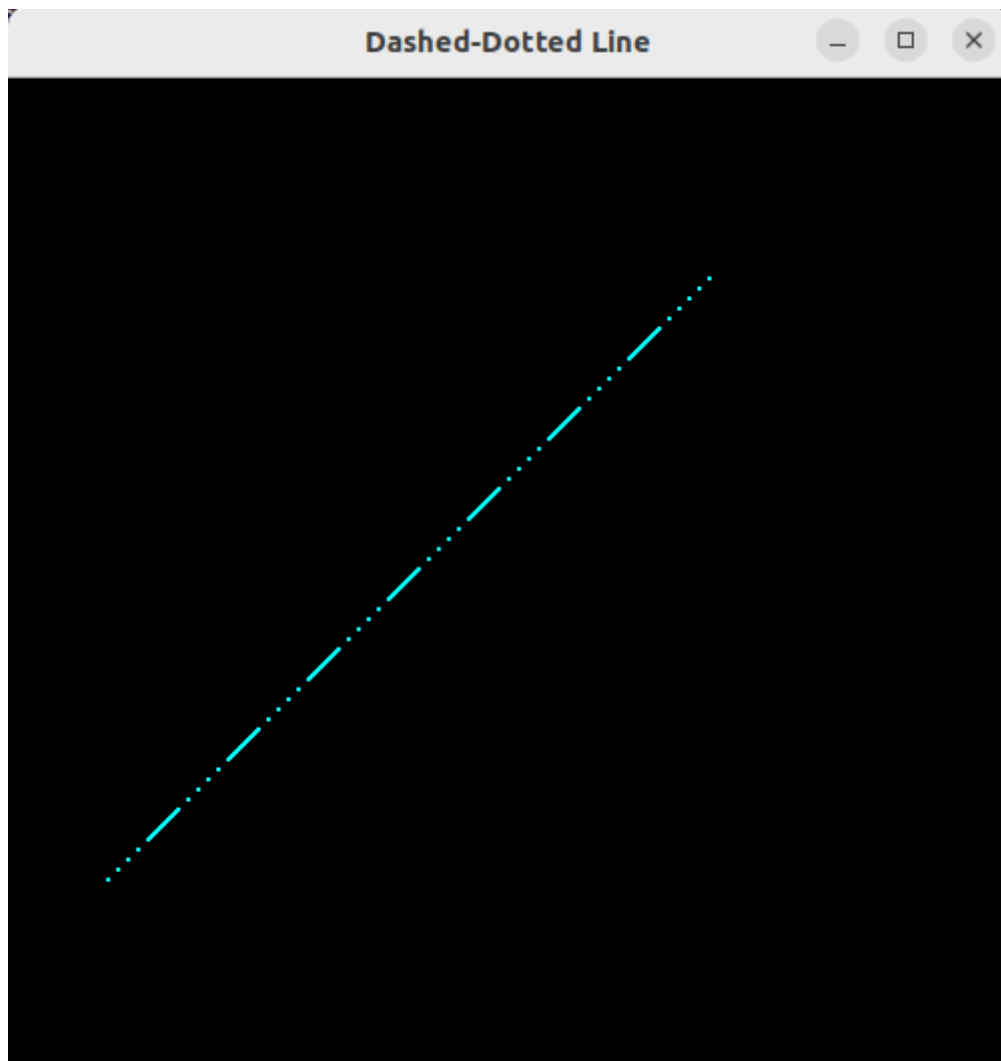
4. Dashed-Dotted Line

```
1 #include <GL/glut.h>
2 #include <iostream>
3 #include <cmath>
4 using namespace std;
5
6 double x1 = 0.0, Y1 = 0.0;
7 double x2 = 0.0, Y2 = 0.0;
8
9 void init()
10 {
11     glClearColor(0.0, 0.0, 0.0, 0.0);
12     glMatrixMode(GL_PROJECTION);
13     gluOrtho2D(0.0, 500.0, 0.0, 500.0);
14 }
15
16 void input()
17 {
18     cout << "Enter the coordinates of the first point: ";
19     cin >> x1 >> Y1;
20     cout << "Enter the coordinates of the second point: ";
21     cin >> x2 >> Y2;
22 }
23
24 void display()
25 {
26     glClear(GL_COLOR_BUFFER_BIT);
27     glColor3f(0.0, 1.0, 1.0);
28     glPointSize(2.0);
29
30     double dy = Y2 - Y1;
31     double dx = x2 - x1;
32     double slope = (dy * 1.0 / dx);
33
34     double p = (2 * dy) - dx;
35     double x = x1;
36     double y = Y1;
37     int count = 0;
38     bool flag = true; // true is for dotted, false is for dashed
39     int dot_counter = 5;
40     int dash_counter = 15;
41
42     glBegin(GL_POINTS);
43     while(x <= x2)
44     {
45         if(dot_counter > 0) {
46             if(count == 0)
47             {
48                 glVertex2i(x, y);
49                 count = 5;
50                 dot_counter -= 1;
51                 if(!dot_counter)
52                     dash_counter = 15;
```

```

54         count--;
55         if(p < 0)
56         {
57             p += 2 * dy;
58         }
59         else
60         {
61             y += 1;
62             p += 2 * dy - 2 * dx;
63         }
64         x += 1;
65     }
66     else
67     {
68         glVertex2i(x, y);
69         dash_counter -= 1;
70         if(p < 0)
71         {
72             p += 2 * dy;
73         }
74         else
75         {
76             y += 1;
77             p += 2 * dy - 2 * dx;
78         }
79         x += 1;
80         if(!dash_counter)
81             dot_counter = 5;
82     }
83 }
84
85 }
86
87 glEnd();
88 glFlush();
89 }
90
91
92 int main(int argc, char **argv)
93 {
94     input();
95     glutInit(&argc, argv);
96     glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
97     glutInitWindowSize(500, 500);
98     glutCreateWindow("Dashed-Dotted Line");
99     init();
100    glutDisplayFunc(display);
101    glutMainLoop();
102    return 0;
103 }
104

```



5. Draw HELLO using lines.

```
1 #include <GL/glut.h>
2
3 void drawH(float x, float y)
4 {
5     glBegin(GL_LINES);
6         glVertex2f(x, y);        glVertex2f(x, y + 0.2);
7         glVertex2f(x + 0.1, y); glVertex2f(x + 0.1, y + 0.2);
8         glVertex2f(x, y + 0.1); glVertex2f(x + 0.1, y + 0.1);
9     glEnd();
10 }
11
12 void drawE(float x, float y)
13 {
14     glBegin(GL_LINES);
15         glVertex2f(x, y);        glVertex2f(x, y + 0.2);
16         glVertex2f(x, y + 0.2); glVertex2f(x + 0.1, y + 0.2);
17         glVertex2f(x, y + 0.1); glVertex2f(x + 0.08, y + 0.1);
18         glVertex2f(x, y);        glVertex2f(x + 0.1, y);
19     glEnd();
20 }
21
22 void drawL(float x, float y)
23 {
24     glBegin(GL_LINES);
25         glVertex2f(x, y);        glVertex2f(x, y + 0.2);
26         glVertex2f(x, y);        glVertex2f(x + 0.1, y);
27     glEnd();
28 }
29
30 void drawO(float x, float y)
31 {
32     glBegin(GL_LINES);
33         glVertex2f(x, y);        glVertex2f(x, y + 0.2);
34         glVertex2f(x, y + 0.2); glVertex2f(x + 0.1, y + 0.2);
35         glVertex2f(x + 0.1, y + 0.2); glVertex2f(x + 0.1, y);
36         glVertex2f(x + 0.1, y);  glVertex2f(x, y);
37     glEnd();
38 }
39
40
```

```

40 void display()
41 {
42     glClear(GL_COLOR_BUFFER_BIT);
43     glColor3f(0.0, 1.0, 1.0);
44
45     float startX = -0.9f;
46     float startY = 0.0f;
47     float gap = 0.15f;
48
49     drawH(startX, startY);
50     drawE(startX + gap, startY);
51     drawL(startX + 2 * gap, startY);
52     drawL(startX + 3 * gap, startY);
53     drawO(startX + 4 * gap, startY);
54
55     glFlush();
56 }
57
58 void init()
59 {
60     glClearColor(0, 0, 0, 1);
61     glMatrixMode(GL_PROJECTION);
62     glLoadIdentity();
63     gluOrtho2D(-1, 1, -1, 1);
64 }
65
66 int main(int argc, char** argv)
67 {
68     glutInit(&argc, argv);
69     glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
70     glutInitWindowSize(600, 400);
71     glutCreateWindow("HELLO using Lines");
72     init();
73     glutDisplayFunc(display);
74     glutMainLoop();
75     return 0;
76 }
77

```

